

The frustum of a pyramid

A pyramid with its top cut off — and the trapezium faces that result.

LESSON 35

1 × 40 MIN

GEOMETRY 11, §16

EXTENSION

LESSON CLOCK

8'

14'

14'

6'

3'

What a frustum is	8 min
Surface area	14 min
Volume	14 min
Recovering the pyramid	6 min
Homework	3 min

LEARNING OBJECTIVES

- Describe the elements of a frustum of a pyramid.
- Compute its lateral and total surface area.
- Compute its volume.
- Recover the original pyramid from the frustum.

TEXTBOOK

National	pp. 173–178
Cambridge	Core & Extended, pp. 331–334
Workbook	Exercise 16.1

01

Explanation

What a frustum is

Cut a pyramid with a plane parallel to its base and remove the small pyramid on top. What is left is a **frustum**. Its two bases are **similar** polygons in parallel planes, and its lateral faces are trapezia.

THE KEY RATIO

If the cut is at height k of the way up, the small pyramid is similar to the whole with scale factor k — so its base edges are k times, its area k^2 times, and its volume k^3 times.

Surface area

$$S_{lat} = \frac{1}{2}(P_1 + P_2) \cdot l$$

where P_1, P_2 are the two base perimeters and l is the slant height of the frustum — the height of one trapezium face. This is the trapezium area formula, once per face, collected.

$$S_{total} = B_1 + B_2 + \frac{1}{2}(P_1 + P_2)l$$

Example. A frustum of a square pyramid with bases 10 and 4, slant height 5: $S_{lat} =$

$$\frac{1}{2}(40 + 16)(5) = 140; S = 100 + 16 + 140 = 256.$$

Volume

$$V = \frac{h}{3}(B_1 + B_2 + \sqrt{B_1 B_2})$$

It is the big pyramid minus the small one, simplified. The middle term $\sqrt{B_1 B_2}$ is the geometric mean of the two bases, and it is what makes the formula more than an average.

Example. Bases 10 and 4 (so $B_1 = 100$, $B_2 = 16$), height 4:

$$V = \frac{4}{3}(100 + 16 + 40) = \frac{4}{3}(156) = 208$$

NOT THE AVERAGE OF THE TWO BASES

$\frac{h}{2}(B_1 + B_2)$ would give 232 — too big. The frustum tapers, and the geometric mean is what accounts for it.

Recovering the pyramid

If the bases are a and b with $b < a$, the scale factor is $k = \frac{b}{a}$. The full pyramid's height H satisfies

$$H - h = kH \Rightarrow H = \frac{h}{1 - k}$$

For bases 10 and 4 with $h = 4$: $k = 0.4$, so $H = \frac{4}{0.6} \approx 6.67$. Check the volume: $\frac{1}{3}$

$$(100)(6.67) - \frac{1}{3}(16)(2.67) = 222 - 14.2 \approx 208 \checkmark.$$

02 Worked examples

EXAMPLE 1 A frustum of a square pyramid has bases 10 and 4 and slant height 5 . Find its total surface area.

$$(1) \quad P_1 = 40, P_2 = 16$$

$$(2) \quad S_{lat} = \frac{1}{2}(56)(5) = 140$$

$$(3) \quad B_1 = 100, B_2 = 16$$

$$(4) \quad S = 256$$

ANSWER

256

EXAMPLE 2 The same frustum has height 4 . Find its volume.

$$(1) \quad \sqrt{B_1 B_2} = \sqrt{1600} = 40$$

$$(2) \quad V = \frac{4}{3}(100 + 16 + 40)$$

$$(3) \quad = \frac{4}{3} \times 156 = 208$$

ANSWER

208

EXAMPLE 3 Find the height of the original pyramid.

$$\textcircled{1} \quad k = \frac{4}{10} = 0.4$$

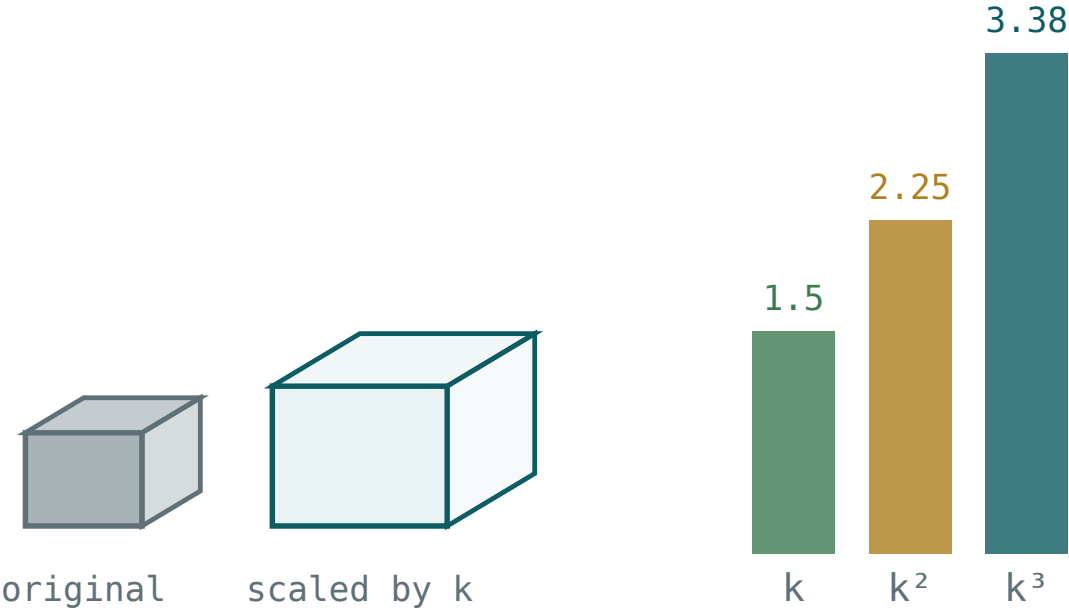
$$\textcircled{2} \quad H = \frac{h}{1 - k} = \frac{4}{0.6}$$

$$\textcircled{3} \quad \approx 6.67$$

ANSWER ≈ 6.67 **03** **Interactive model**

Cut a card pyramid parallel to its base and hold up both pieces.

Cutting a pyramid



k 1.5	length × 1.5	area ×k² 2.25	volume ×k³ 3.38
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04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Frustum	Kesik piramida	Усечённая пирамида
Upper base	Yuqori asos	Верхнее основание
Lower base	Quyi asos	Нижнее основание
Slant height of a frustum	Kesik piramida apofemasi	Апофема усечённой пирамиды
Lateral face	Yon yoq	Боковая грань
Similar bases	O'xshash asoslar	Подобные основания
Scale factor	O'xshashlik koeffitsienti	Коэффициент подобия
Complementary cone	To'ldiruvchi piramida	Дополняющая пирамида

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

The lateral faces of a frustum are:

triangles

trapezia

rectangles

parallelograms

The two bases of a frustum are:

congruent

similar

unrelated

perpendicular

S_{lat} of a frustum:

$$\frac{1}{2}Pl$$

$$\frac{1}{2}(P_1 + P_2)l$$

$$(P_1 + P_2)l$$

$$Pl$$

V of a frustum:

$$\frac{h}{2}(B_1 + B_2)$$

$$\frac{h}{3}(B_1 + B_2 + \sqrt{B_1 B_2})$$

$$\frac{1}{3} Bh$$

$$Bh$$

The average-of-bases formula gives an answer that is:

correct

too big

too small

zero

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. S_{lat} with $P_1 = 40$, $P_2 = 16$, $l = 5$

ANSWER 140

2. S_{lat} with $P_1 = 24$, $P_2 = 12$, $l = 4$

ANSWER 72

3. $\sqrt{B_1 B_2}$ for $B_1 = 100$, $B_2 = 16$

ANSWER 40

4. $\sqrt{B_1 B_2}$ for $B_1 = 36$, $B_2 = 9$

ANSWER 18

5. The lateral faces of a frustum are

ANSWER trapezia

6. The bases of a frustum are

ANSWER similar

7. Scale factor for bases 4 and 10

ANSWER 0.4

Medium · the standard question

7 PROBLEMS

1. Bases 10 and 4, slant height 5: total area

ANSWER 256

2. Bases 10 and 4, height 4: the volume

ANSWER 208

3. Height of the original pyramid there

ANSWER ≈ 6.67

4. Bases 6 and 3, height 4: the volume

ANSWER 84

5. Bases 12 and 6, slant height 8: lateral area

ANSWER 288

6. Bases 8 and 4, height 6: the volume

ANSWER 224

7. A frustum cut halfway up: the volume ratio to the whole

ANSWER 7 : 8

Hard · stretch and reason

7 PROBLEMS

1. Bases 9 and 6, slant height 5: total area and volume with $h = 4$

ANSWER $S = 267$; $V \approx 254$

2. A frustum of volume 208 with bases 100 and 16: its height

ANSWER 4

3. Prove $V = \frac{h}{3}(B_1 + B_2 + \sqrt{B_1 B_2})$

ANSWER Subtract the small pyramid from the whole

4. A frustum has bases in ratio 1 : 2. The volume ratio of the frustum to the whole pyramid

ANSWER 7 : 8

5. A bucket: bases 24 and 18 cm, depth 20 cm. Capacity in litres

ANSWER ≈ 6.9 litres

6. A frustum with equal bases is what solid?

ANSWER a prism

7. Show the average formula overestimates

ANSWER $\frac{B_1+B_2}{2} > \frac{B_1+B_2+\sqrt{B_1 B_2}}{3}$ by AM–GM

07 Homework

Homework — 4 tasks

Never average the two bases — use the geometric-mean formula.

1. A frustum of a square pyramid has bases 12 and 6 and slant height 7. Find its total surface area.
2. The same frustum has height 6. Find its volume.
3. Find the height of the pyramid it was cut from.

4. Explain in three sentences why the volume is not $\frac{h}{2}(B_1 + B_2)$.